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* رقم الكمبيوتر

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 Will be reviewed

8

What is the primary purpose of 'Aging' in priority-based scheduling algorithms? *

☒ To prevent starvation of low-priority processes.

☐ To decrease the overhead of context switching.

☐ To reduce the turnaround time of short jobs.

☐ To allocate more memory to older processes.

 Will be reviewed

Consider the following snapshot of a system; If a request from process P1 arrives for (0, 0, 0, 1), can the request be granted immediately?

★

	<u>Allocation</u>				<u>Max</u>				<u>Available</u>			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	2	3	1	4	3	5	2	4	2	0	3	1
P1	1	1	0	2	2	1	3	4				
P2	0	1	2	1	2	2	2	6				
P3	0	0	2	1	2	0	3	2				
P4	2	0	1	3	5	1	1	3				

☐ Yes

☒ No

📝 Will be reviewed

10

Which scheduling algorithm is susceptible to the 'Convoy Effect'? ★

☐ Priority Scheduling

☐ Round Robin

☒ First-Come First-Served (FCFS)

☐ Shortest Job First (Preemptive)

📝 Will be reviewed

11

When a process is in the 'Running' state and its allocated time quantum expires, which state does it transition to? ★



☐ Ready

☐ New

☒ Waiting

☐ Terminated

 **Will be reviewed**

12

What is the relationship between a 'Safe State' and a 'Deadlock' in a system? *

☐ Deadlock can only occur when the system is in a safe state.

☐ Every unsafe state is a deadlock state.

☒ A safe state is never a deadlock state, but an unsafe state may or may not lead to a deadlock.

☐ A safe state is one where no processes are currently holding any resources.

 **Will be reviewed**

13

What is the primary motivation for using a 'File Allocation Table' (FAT) instead of standard Linked Allocation? *

☐ To eliminate the need for a directory structure.

☐ To encrypt the file data automatically.

☐ To allow multiple processes to write to the same block simultaneously.

☒ To improve random access speed by keeping all pointers in a cached table.

 Will be reviewed

14

Which CPU scheduling algorithm is designed specifically for time-sharing systems to provide a fair distribution of CPU time among processes? *

- ☐ Shortest Remaining Time First (SRTF)
- ☐ Non-preemptive Priority Scheduling
- ☐ First-Come First-Served (FCFS)
- ☒ Round Robin (RR)

 Will be reviewed

15

Which hardware component is responsible for dynamically mapping logical addresses used by a process to physical addresses in memory? *

- ☐ Arithmetic Logic Unit (ALU)
- ☐ Translation Unit
- ☐ Input/Output Controller
- ☒ Memory Management Unit (MMU)

 Will be reviewed

16

The Banker's Algorithm is used for deadlock avoidance. What information must a process provide when it first enters the system for this algorithm to work? *

- ☐ A list of all other processes it will communicate with.
- ☐ The priority level assigned to it by the CPU scheduler.
- ☐ The exact time at which it will finish execution.
- ☒ The maximum number of instances of each resource type it may ever need.

 **Will be reviewed**

17

Which of the following data structures is used by the operating system to store all the information needed to manage a specific process? *

- ☐ Ready Queue
- ☐ File Descriptor Table
- ☒ Process Control Block
- ☐ Segment Table

 **Will be reviewed**

18

In CPU scheduling, how is 'Turnaround Time' defined? *

- ☒ The interval from the time of submission of a process to the time of completion.
- ☐ The time it takes from the first request until the first response is produced.
- ☐ The number of processes that complete their execution per unit time.
- ☐ The time a process spends waiting in the ready queue.

📝 Will be reviewed

19

Consider the following set of processes, with the length of the CPU burst given in milliseconds. The processes are assumed to have arrived all at time 0. What is the turnaround time of process P2 using Round Robin (RR, quantum = 2 millisec) algorithm?

*

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

☐ 7

☒ 3

☐ 1

☐ 12

📝 Will be reviewed

20

Which of the following is a significant disadvantage of Contiguous Allocation? *

☐ External Fragmentation

☐ Slow random access speed

☒ High pointer overhead within data blocks

☐ Complex directory structures

 **Will be reviewed**

21

Which of the following describes a 'Race Condition'? *

☐ The speed at which the CPU context switches between two different threads.

☐ A situation where two processes are waiting for each other to release a resource.

☒ Multiple processes access and manipulate shared data concurrently, and the outcome depends on the order of execution.

☐ A process consuming of the CPU cycles.

 **Will be reviewed**

22

What is the primary conceptual advantage of using Virtual Memory in a multi-programming environment? *

☐ It prevents the use of page tables for address translation.

☐ It eliminates the need for any physical RAM in the system.

☒ It allows the logical address space to be larger than the physical address space.

☐ It speeds up the CPU clock cycle during context switches.

 **Will be reviewed**

23

What is the 'Critical Section' in the context of process synchronization? *

- ☐ The memory area reserved for the operating system kernel.
- ☐ The part of the PCB that stores the process ID.
- ☒ A segment of code where shared resources are accessed.
- ☐ A state where a process is waiting for I/O completion.

 Will be reviewed

24

In a paged memory system, if the logical address space is 2^{32} bytes and the page size is 4 KB (2^{12} bytes), how many entries will there be in a single-level page table? *

- ☐ 2^{44}
- ☐ 2^{12}
- ☐ 2^{20}
- ☒ 2^{32}

 Will be reviewed

25

Consider the following page-reference strings: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6
How many page faults would occur for the Optimal replacement algorithm, assuming four memory frames? *

- ☐ 12

☒ 8

☐ 14

☐ 10

 **Will be reviewed**

26

Consider the following page-reference strings: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6
How many page faults would occur for the FIFO replacement algorithm, assuming four memory frames? *

☒ 14

☐ 12

☐ 10

☐ 8

 **Will be reviewed**

27

Consider the following set of processes, with the length of the CPU burst given in milliseconds. The processes are assumed to have arrived all at time 0. What is the turnaround time of process P3 using Round Robin (RR, quantum = 2 millisec) algorithm? *

Process Burst Time

P1 6

P2 1

P3 3

P4 2

P5 5

☐ 1

☐ 7

☒ 12

☐ 3

 Will be reviewed

28

In a demand-paging system, what is the 'Valid-Invalid' bit in the page table used for? *

☐ To indicate the priority level of the process owning the page.

☐ To determine if a page has been modified since it was loaded.

☒ To distinguish between pages currently in physical memory and those on the disk.

☐ To count how many times a page has been accessed by the CPU.

 Will be reviewed

How does an 'Indexed Access' method typically locate a specific record in a large file? *

- ☐ By using a mathematical hash of the entire file contents.
- ☐ By reading every block from the start until the record is found.
- ☐ By moving the disk head to a random position and checking for the data.
- ☒ By locating small index block first to find a pointer to the actual data.

 **Will be reviewed**

What is the primary use of binary semaphores? *

- ☒ To implement mutual exclusion.
- ☐ To manage memory allocation.
- ☐ To count the number of available resources.
- ☐ To schedule processes.

 **Will be reviewed**

Consider the following snapshot of a system; Is the system in a safe state?

*

	<u>Allocation</u>				<u>Max</u>				<u>Available</u>			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	2	3	1	4	3	5	2	4	2	0	3	1
P1	1	1	0	2	2	1	3	4				
P2	0	1	2	1	2	2	2	6				
P3	0	0	2	1	2	0	3	2				
P4	2	0	1	3	5	1	1	3				

☐ No

☒ Yes

 **Will be reviewed**

32

What is the primary disadvantage of the Shortest Seek Time First (SSTF) scheduling algorithm? *

☐ It is extremely complex to implement in modern operating systems.

☒ It can cause starvation for requests that are far from the current head position.

☐ It always results in the maximum possible head movement.

☐ It requires the disk to rotate at a variable speed.

 **Will be reviewed**

33

In Demand Paging, what occurs immediately after a process attempts to access a page that is marked as 'invalid' in the page table? *

☐ The MMU corrects the address automatically without the OS.

☐ The process is terminated immediately.

- ☒ A Page Fault trap is sent to the operating system.
- ☐ The CPU stalls until the user provides more RAM.

 **Will be reviewed**

34

In a Resource Allocation Graph, what does a directed edge from a Process to a Resource represent? *

- ☒ The process is requesting an instance of the resource.
- ☐ The process has finished using the resource.
- ☐ The resource is currently allocated to the process.
- ☐ The resource is blocked by a deadlock.

 **Will be reviewed**

35

Consider the following page-reference strings: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6
How many page faults would occur for the LRU replacement algorithm, assuming four memory frames? *

- ☐ 12
- ☐ 8
- ☒ 10
- ☐ 14

 Will be reviewed

36

Which page replacement algorithm is based on the idea of replacing the page that will not be used for the longest period of time in the FUTURE? *

- ☒ Optimal Algorithm
- ☐ Least Frequently Used (LFU)
- ☐ Least Recently Used (LRU)
- ☐ First-In First-Out (FIFO)

 Will be reviewed

37

Unlike paging, which divides memory into fixed-size blocks, how does 'Segmentation' divide a process's logical address space? *

- ☒ Into variable-sized blocks based on logical units like functions or data arrays.
- ☐ Into blocks that are always a multiple of the sector size on disk.
- ☐ Into fixed-size blocks.
- ☐ Into one single contiguous block of physical memory.

 Will be reviewed

38

What is the key optimization that the LOOK algorithm provides over the SCAN algorithm? *

- ☐ It predicts future requests based on past user behavior.
- ☐ It reduces the rotational latency by speeding up the platter.
- ☐ It ensures that no two requests are ever serviced in the same track.
- ☒ The arm only goes as far as the last request in each direction instead of traveling to the very end of the disk.

 **Will be reviewed**

39

Which of the following is NOT one of the four necessary conditions for a deadlock to occur in a system? *

- ☐ Circular Wait
- ☒ Preemption
- ☐ Mutual Exclusion
- ☐ Hold and Wait

 **Will be reviewed**

40

Which memory allocation strategy searches the entire list of free blocks and allocates the smallest block that is big enough for the request? *

- ☐ Next-fit
- ☐ First-fit
- ☐ Worst-fit

☒ Best-fit

 **Will be reviewed**

41

A disk queue has requests for cylinders [98,183,41]. If the head is currently at cylinder 53 and using FCFS, what is the total head movement in cylinders? *

☒ 272

☐ 142

☐ 130

☐ 322

 **Will be reviewed**

42

What is the primary objective of 'Context Switching' in a multi-tasking operating system? *

☐ To increase the total execution speed of a single program.

☐ To terminate a process that is consuming too many resources.

☐ To allocate more memory to a high-priority process.

☒ To save the state of the current process and load the state of the next process.

 **Will be reviewed**

43

In a Round Robin scheduling system with a time quantum $q=4\text{ms}$, how much time will a process with a burst time of 10ms spend in the 'Running' state during its first turn? *

- ☐ 10ms
- ☐ 2.5ms
- ☐ 6ms
- ☒ 4ms

 **Will be reviewed**

44

Which of the following phenomena describes a situation where increasing the number of page frames results in an INCREASE in the number of page faults? *

- ☐ Segmentation Fault
- ☐ Thrashing
- ☒ Belady's Anomaly
- ☐ The Locality Principle

 **Will be reviewed**

45

What are the two primary pieces of information stored in the directory entry for a file using Contiguous Allocation? *

- ☒ The starting address and a pointer to the File Allocation Table

- ☒ Starting address and length of the file
- ☐ A list of all block numbers used by the file
- ☐ Starting address and ending address pointers

📝 **Will be reviewed**

46

What is the main difference between 'Shared Memory' and 'Message Passing' in Inter-Process Communication (IPC)? *

- ☐ Shared memory is managed entirely by the hardware without OS intervention.
- ☐ Message passing is less secure than shared memory.
- ☐ Message passing is only possible between processes on different computers.
- ☒ Shared memory is generally faster as it avoids system calls for every data transfer.

📝 **Will be reviewed**

47

Which of the following is a characteristic of a 'Short-term Scheduler'? *

- ☒ It selects from among the processes in the ready queue and allocates the CPU to one of them.
- ☐ It removes processes from memory temporarily to reduce the degree of multiprogramming.
- ☐ It selects processes from a pool and loads them into memory for execution.
- ☐ It manages the communication between two different networked computers.

 Will be reviewed

48

How does C-SCAN differ from the standard SCAN algorithm? *

- ☐ It prioritizes requests based on the file size rather than the cylinder number.
- ☐ It uses two disk arms simultaneously to increase throughput.
- ☐ It only services the requests at the very center of the disk.
- ☒ After reaching one end, the head immediately returns to the beginning without servicing any requests on the return trip.

 Will be reviewed

49

In the standard process state transition diagram, which transition occurs when the CPU scheduler picks a process from the 'Ready' queue to execute? *

- ☐ Waiting to Ready
- ☐ Running to Ready
- ☐ New to Ready
- ☒ Ready to Running

 Will be reviewed

50

What is the primary purpose of the Test and Set instruction? *

- ☐ To schedule processes.

- ☐ To manage memory allocation.
- ☐ To handle I/O requests.
- ☒ To provide a hardware-based solution for mutual exclusion.

📝 **Will be reviewed**

51

If a memory allocation system uses a fixed-partition scheme and a process is assigned a 16 KB partition but only uses 10 KB, what is the resulting 6 KB of unused space called? *

- ☐ Segmentation
- ☒ Internal Fragmentation
- ☐ External Fragmentation
- ☐ Compaction

📝 **Will be reviewed**

52

What is the primary purpose of a Translation Lookaside Buffer (TLB)? *

- ☐ To prevent unauthorized processes from accessing kernel memory.
- ☒ To speed up the translation of virtual addresses to physical addresses.
- ☐ To manage the transfer of data between RAM and Disk.
- ☐ To store the actual data being processed by the CPU.

 Will be reviewed

53

In the context of Deadlock Prevention, how is the 'Circular Wait' condition typically negated? *

- ☒ By requiring processes to request all needed resources at the beginning of execution.
- ☐ By imposing a total ordering of all resource types and requiring processes to request resources in increasing order.
- ☐ By making all resources sharable among all processes.
- ☐ By allowing the OS to take resources away from a process if its next request cannot be immediately granted.

 Will be reviewed

54

Which component of disk access time is the most significant and is the primary target for optimization by disk scheduling algorithms? *

- ☒ Seek Time
- ☐ Transfer Time
- ☐ Rotational Latency
- ☐ Queueing Delay

 Will be reviewed

55

In the Dining-Philosophers Problem, what do the "chopsticks" represent? *

- ☒ Shared resources.
- ☐ Process states.
- ☐ Memory addresses.
- ☐ I/O requests.

📝 Will be reviewed

56

Consider the following set of processes, with the length of the CPU burst given in milliseconds. The processes are assumed to have arrived all at time 0. What is the waiting time for process P3 using Round Robin (RR, quantum = 2 millisec) algorithm? *

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

- ☒ 9
- ☐ 12
- ☐ 5
- ☐ 2

57

Consider the following set of processes, with the length of the CPU burst given in milliseconds. The processes are assumed to have arrived all at time 0. What is the waiting time for process P3 using Shortest Job First (SJF) algorithm? *

Process Burst Time

P1	6
P2	1
P3	3
P4	2
P5	5

☐ 1

☒ 3

☐ 6

☐ 0



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